

I $\rightarrow$ decision is, H_0 's negaⁿ

II $\rightarrow$ decision is, H_0

Practice for Quiz II

1. H_0 : loan app.
 H_1 : not app.

H_0 : not app.
 H_1 : app.

(a) type I $\Rightarrow$ app.

$\hookrightarrow$ (reject H_0 | H_0 is true)

H_0 : should not be approved

H_1 : should be approved

(b) type II $\Rightarrow$ app.

$\hookrightarrow$ reject (fail to reject H_0 | H_0 is false)

H_0 : should be approved

H_1 : should not be approved

2. H_0 : $p = 0.4$

H_1 : $p < 0.4$ ($X \leq 3$)

(a) (reject H_0 | H_0 is true)

~~$P(X \leq 0.4 | p = 0.4)$~~

$$\alpha = P(X \leq 3 | p = 0.4) = \sum_{x=0}^3 \binom{18}{x} (0.4)^x (0.6)^{18-x}$$
$$= 0.0327$$

$$\alpha \approx 3.28\%$$

(b) (fail to reject H_0 | H_0 is false)

$$\beta = P(X \geq 4 | p) \Rightarrow 1 - P(X \leq 3 | p)$$

$$0.2 \Rightarrow 0.4990$$

$$0.1 \Rightarrow 0.0982$$

3. $\sigma = 6$, $n = 36$

H_0 : $\mu = 50$, H_1 : $\mu \neq 50$

reject H_0 : $|\bar{X} - 50| > 2.5$

(a) type-I $\Rightarrow$ (reject H_0 | H_0 is true)

$$\bar{X} \sim N(50, 1^2)$$

reject $\bar{X} < 47.5$ or $\bar{X} > 52.5$

$$\alpha = P(|Z| > 2.5) = 2P(Z > 2.5)$$

$$\phi(2.5) \approx 0.99379 \Rightarrow 0.0124$$

(b) type-II $\Rightarrow \mu = 52$ $\bar{X} \sim N(52, 1)$
fail to reject $47.5 \leq \bar{X} \leq 52.5$

$$\beta = P(47.5 \leq \bar{X} \leq 52.5 \mid \mu = 52)$$

$$Z_L = \frac{47.5 - 52}{1}$$

$$= -4.5$$

$$Z_U = \frac{52.5 - 52}{1} = 0.5$$

$$\beta = \phi(0.5) - \underbrace{\phi(-4.5)}_0$$

$$= 0.6915$$

(c) $n = ?$, $\alpha = 0.05$, power ≥ 0.9 at $\mu = 52$

$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$\text{Power} = 0.9, \quad \beta = 1 - \text{Power} = 0.1 \quad Z_{0.1} = 1.2816$$

$$\text{effect} \Rightarrow |\mu - \mu_0| = |52 - 50| = 2$$

$$n \approx \left[\frac{(Z_{\alpha/2} + Z_{\beta}) \sigma^2}{\mu - \mu_0} \right] \cdot \left[\frac{((1.96 + 1.2816)2^2)}{2} \right]^2$$

$$= 94.57$$

$$= 95$$

4. $n = 20$, $\bar{x} = 8.6$, $s = 1.2$

$H_0: \mu = 8$, $H_1: \mu > 8$, $\alpha = 0.05$ $t_{0.05, 19} = 1.729$

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{8.6 - 8}{1.2/\sqrt{20}} = 2.236$$

$2.236 > 1.729$ reject $H_0 \Rightarrow \mu > 8$

5. $n_1 = 10, \bar{x}_1 = 42.5, s_1^2 = 6.2$
 $n_2 = 12, \bar{x}_2 = 39.8, s_2^2 = 5.4$

$H_0: \mu_1 = \mu_2$

$v = n_1 + n_2 - 2$

$H_1: \mu_1 \neq \mu_2$

$= 20$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(9)(6.2) + (12)(5.4)}{20} = 5.76$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = 2.627$$

$t_{0.025, 20} = 2.086$

$|2.627| > 2.086$

$\Rightarrow$ reject H_0

$\Rightarrow \mu_1 \neq \mu_2$

6. $\bar{d} = 2.3, s_d = 3.1$

$\alpha = 0.05$

$H_0: \mu_d = 0$

$t_{0.05, 9} = 1.833$

$H_1: \mu_d > 0$

$$t = \frac{2.3 - 0}{3.1 / \sqrt{10}} = \frac{2.3}{3.1 / \sqrt{10}} = 2.346$$

$2.346 > 1.833 \Rightarrow$ reject $H_0 \Rightarrow \mu_d > 0$

because the before & after obs. belong to the same subjects, obs. are paired & $\therefore$ are not independent.

7. $H_0: \sigma^2 \leq 8$

$n = 25, s^2 = 12.4$

$H_1: \sigma^2 > 8$

$\alpha = 0.05, \chi^2_{0.05, 24} = 36.415$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(24)(12.4)}{8} = 37.2$$

$\chi^2_{calc} > \chi^2_{obs} \Rightarrow$ reject $H_0 \Rightarrow \sigma^2 > 8$

$$37.2 - 36.415 = 0.785 \Rightarrow \text{borderline "reject"}$$

8. $n=15, s^2 = 0.36, \alpha = 0.05$

(a) $H_0: \sigma^2 = 0.25$ $\chi^2_{0.025, 14} = 26.119$

$H_1: \sigma^2 \neq 0.25$

$\chi^2_{0.975, 14} = 5.629$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(14)(0.36)}{0.25} = 20.16$$

$$5.629 < 20.16 < 26.119 \Rightarrow \text{fail to reject } H_0$$

(b) $\frac{(14)(0.36)}{26.119} < \sigma^2 < \frac{(14)(0.36)}{5.629}$

$$(0.193, 0.895) \Rightarrow 95\% \text{ CI}$$

(c) $0.25 \in (0.193, 0.895)$, also like the we fail to reject H_0

9. $\alpha = 0.05, s_1^2 = 6.2, s_2^2 = 5.4$

$H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

$$F = \frac{s_1^2}{s_2^2} = 1.148$$

$F_{0.025, 9, 11} = 3.588$

$F_{0.975, 9, 11} = 0.256$

$$0.256 < 1.148 < 3.588$$

$$\Rightarrow \text{fail to reject } H_0$$

10. $n=120, 15, 23, 18, 25, 21, 18, \alpha = 0.05$

$\chi^2_{0.05, 5} = 11.070$

$p_i = \frac{1}{6}$

$E_i = \frac{1}{6} \times 120 = 20$

$H_0: p_1 = \dots = p_6 = \frac{1}{6}$

$H_1: \text{at least 1 differs}$

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$$

$\chi^2 = 3.40$

$3.40 < 11.070$

fail to reject H_0

11. $n = 4$, $9:3:3:1$ $n = 160$
 $\alpha = 0.05$, $\chi^2_{0.05,3} = 7.815$ 95, 28, 24, 13

$H_0: p_1 = \frac{9}{16}, p_2 = \frac{3}{16}, p_3 = \frac{3}{16}, p_4 = \frac{1}{16}$

$H_1: \text{any 1 differs}$ 90:30:30:10

$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 2.511$

$2.511 < 7.815 \Rightarrow \text{fail to reject } H_0$

12.

	D	A	
M	20	180	200
A	30	170	200
N	50	150	200
	100	500	600

$\alpha = 0.05$
obs.

$\chi^2_{0.05,2} = 5.991$

$H_0: \text{shift \& quality are indep.}$
 $H_1: \text{not indep.}$

def. $E_{ij} = \frac{200 \times 100}{600} = 33.333$

$E_{ij}(\text{accep}) = \frac{200 \times 500}{600} = 166.667$

	D	A	
M	33	166	
A	33	166	
N	33	166	16.80

$\chi^2 = \sum \frac{(O_i - E_{ij})^2}{E}$

$v = (r-1)(c-1) = 1 \times 2 = 2$

$16.80 > 5.991 \Rightarrow \text{reject } H_0$
 not indep.
 night

13. $n=15$, $\text{med} = 4200$, $B = 2000$
 $B_{50} = 3150$, $B_{1950} = 5600$

(a) $\alpha = 0.05$ $(0.025)(2000) = 50$
 $(0.975)(2000) = 1950$
 $CI \Rightarrow (50, 1950)$

(b) it approximates the sample distribⁿ of the sample median by repeatedly sampling from empirical distribⁿ by the obs. sample.
 Not an estimate of the distribⁿ of the indi. repair costs.

(c) $\text{Var}(\bar{X}) \approx \frac{1}{4n f(m)^2}$ it would req. distributional assumⁿ or an estimate of the density near populaⁿ median

(d) $B : 2000 \rightarrow 50\% \Rightarrow CI : (3190, 5570)$

(i) MC error : increasing B , reduces MC error

(ii) bootstrap approximaⁿ error

$B \uparrow$, gives more precise bootstrap computaⁿ

14. $n = 20$, 7 is success, $p = \frac{7}{20} =$
 param $p \sim \text{Beta}(2, 2)$

(a) $X|p \sim \text{Bin}(n, p)$ $p \sim \text{Beta}(a, b)$

param $(p|X) = x \sim \text{Beta}(a+x, b+n-x)$
 $a=2, b=2, x=7, n-x=13$

$p|\text{data} \sim \text{Beta}(2+7, 2+13)$

$\sim \text{Beta}(9, 15)$

(b) Beta (a, b) $E[p] = \frac{a}{a+b}$

$E[p | \text{data}] = \frac{9}{24} = 0.375$

Mode = $\frac{a-1}{a+b-2} = \frac{8}{22} = 0.3636$

$\hat{p}_{MLE} = \frac{7}{20} = 0.35$

(c) 95% credI. $[0.197, 0.573]$

$P(0.197 < p < 0.573 | \text{data}) = 0.95$

(d) if $n \downarrow$, our prior is concentrated then it affects interval

15. $\hat{p}_1 = \frac{120}{300}$ $\hat{p}_2 = \frac{90}{280}$ $\alpha = 50\%$

(a) $H_0: p_1 = p_2$ $\hat{p} = \frac{120+90}{300+280} = 0.362$
 $H_1: p_1 \neq p_2$

(b) $Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ $Z = 1.967$

$|Z| > 1.96$

$|1.967| > 1.96$

reject H_0

(c) under H_0 , $p_1 = p_2$ both populations share 1 common population proportion p , so the null SE should estimate that common p using all obs. together.

16. $\hat{p}_1 = \frac{45}{500}$ $\hat{p}_2 = \frac{28}{450}$ $\alpha = 0.05$

$H_0: p_1 = p_2$

$H_1: p_1 \neq p_2$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = 0.076 \quad Z = -1.605$$

$$Z < -1.645 \quad \text{but } -1.605 > -1.645$$

fail to reject H_0

17. (a) $\hat{p}_1 - \hat{p}_2 = 0.07857$

$$95\% \text{ CI} \Rightarrow (\hat{p}_1 - \hat{p}_2) \pm 1.96 \text{ SE}$$

$$\Rightarrow (0.001, 0.156)$$

(b) yes, because $\hat{p}_1 - \hat{p}_2 > 0$

(c) HT, $p_1 = p_2$, we assume a common p , hence pooling, CI, when estimating $\hat{p}_1 - \hat{p}_2$ we do not assume they are equal.

18. (a) $n_1 = 12, \bar{x}_1 = 85.4, S_1^2 = 24.5$

$n_2 = 15, \bar{x}_2 = 79.1, S_2^2 = 61.3$

$H_0: \sigma_1^2 = \sigma_2^2 \quad \alpha = 0.05$

$H_1: \sigma_1^2 \neq \sigma_2^2 \quad F_{0.025, 14, 11} = 3.359$

$$F = \frac{S_2^2}{S_1^2} = \frac{61.3}{24.5} = 2.502$$

$$2.502 < 3.359 \Rightarrow \text{fail to reject } H_0$$

(b)
$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}} \approx 23.89 \approx 24$$

$H_0: \mu_1 = \mu_2 \quad t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \approx 2.545$

$H_1: \mu_1 \neq \mu_2$

$$2.545 > 2.064 \Rightarrow \text{reject } H_0$$

(c) $sp^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = 45.108$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{sp^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \approx 2.422$$

$\Rightarrow \text{reject } H_0$

19. $n_1 = 150, \bar{x}_1 = 3.42, s_1 = 0.85$
 $n_2 = 180, \bar{x}_2 = 3.19, s_2 = 0.92$

(a) $H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 \neq \mu_2$ $\alpha = 0.05$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{0.23}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = 2.357$$

$$|2.357| > 1.96 \Rightarrow \text{reject } H_0$$

(b) $(\mu_1 - \mu_2) \pm 1.96(SE)$

$$0.23 \pm 0.19123 \Rightarrow (0.0388, 0.4212)$$

(c) procedure is currently reasonable as both the sample sizes are large. By CLT, the sampling distributions are normal even if the original obs. are skewed. if $n_1 = 10, n_2 = 12$, strong skewness would become a serious concern.

20	O_i	P	NP	
	D	38	62	100
	ND	57	43	100
		95	105	200

$$\alpha = 0.05$$

$H_0: P \& D$ are indep.

$H_1: \text{not indep.}$

$$E_{ij} = \frac{100 \times 95}{200} = 47.5$$

$$E_{ij} = \frac{100 \times 105}{200} = 52.5$$

$$E_i \quad P \quad NP \quad \chi^2 = 7.238$$

$$D \quad 47.5 \quad 52.5 \quad (n-1)(c-1) = 1$$

$$ND \quad 47.5 \quad 52.5 \quad \chi^2_{0.05, 1} = 3.841$$

$$7.238 > 3.841 \Rightarrow \text{reject } H_0$$

21. D_i	2	P	C	
<25	52	38	30	120
25-40	28	42	35	105
>40	20	30	25	75
	100	110	90	300

(a) E_i	2	P	C	
<25	40	44	36	
25-40	35	38.5	31.5	
>40	25	27.5	22.5	

H_0 : age group & mode are indep.

H_1 : not indep.

$$(b) \chi^2 = \sum \frac{(O - E)^2}{E} = 9.030$$

$$\chi^2_{0.05, 4} = 9.488$$

$$(n-1)(c-1) = 3 \times 2 = 4$$

$9.030 < 9.488 \Rightarrow$ fail to reject H_0 .

(d) $E_{ij} \geq 5$, our smallest is 22.5. The chi-sq. approximation is clearly adequate.